

**Pre-specified analysis plan
(commit 3fcf8d1; before any data analysis)**

Stage 1 · AD
Leng snRNA-seq, entorhinal cortex
RORB+ vulnerable neurons (s1/s2/s4)
10 donors; donor level, age covariate

Stage 2 · GBM
Neftel Smart-seq2, adult malignant cells
NPC/OPC-like vs AC/MES-like
20 tumours; tumour-blocked design

Stage 3 · H2 test (3,857 shared genes)
RRHO2 (DU quadrant) · hypergeometric (top 200)
· permutation (1,000)

H2 NOT SUPPORTED
0/3 tests in all variants; no overlap
detected with 3 donors per group

Stage 4 · H1 test
AD meta-cohort: GSE48350 + GSE36980
hippocampus, n = 51 (ComBat)
GBM: TCGA-GBM (152) + GTEx (186)

Model A: unadjusted
Model B: + BRETIGEA · Model C: + MuSiC

Shared DEGs
(FDR < 0.05 in both cohorts, same direction)

A: 1,458 → B: 0 · C: 0
ratio_B = ratio_C = 0.000
(threshold: ≤ 0.20 support; ≥ 0.50 reject)

*Exploratory (not used for decisions):
H1 replication (GSE125583) · negative control
pseudoreplication · donor × age*

H1 SUPPORTED
(all sensitivity analyses)

Plan section 12 → Scenario B
Shared bulk DEGs markedly reduced after composition adjustment;
no cell-type-level overlap detected with the available donors